

Natural Misère Quotients: Exact Octal Certificates, Heap Normal Forms, and Grundy’s Game

GPT 5.6 Sol

a9lim

Abstract

Siegel’s valid-transition-table theorem characterizes abstract finite misère quotients. Natural realization asks for more: one uniform heap rule must generate the table at every heap size. We express this homogeneity as an exact quotient-valued trace recurrence. A least-rank separation argument proves that the option-value set determines its value in every reduced valid table. Every value in such a table also has a context reaching the P -portion, so no transition value can occur among its own options. Hence a fixed no-split code with finite quotient has an ultimately periodic pretending function and a finite exact verification procedure. Splitting introduces an unbounded convolution, but an asserted periodic split trace still has a finite certificate. We identify the convolution exactly with a system of unary Boolean language equations. Without the valid-table axioms, totality is Π_1^0 -complete already for the single fixed code 0.7; thus finiteness and commutativity alone cannot yield an automaton argument.

We also prove a comparison theorem: every nontrivial finite quotient has an exact one-species heap realization with finitely many source-local exceptions and the uniform tail move $H_n \rightarrow H_{n-1}$. This is not an octal realization; indeed its inert pad cannot be retained in a verbatim octal encoding once the quotient has more than two elements. For every $n \geq 2$, the code with 2^{n-1} consecutive digits equal to 3 has quotient $\mathcal{T}_n$, the tame quotient of order $2^n + 2$; the proof gives the quotient map and the outcome strategy for arbitrary heap multiplicities. For misère Grundy’s game, the quotient generated by heaps through 12 is $\mathcal{T}_2$, heap 13 enlarges it to a reduced monoid of order 12, and heap 18 enlarges it to one of order 24. These are exact submonoid theorems, not bounded indistinguishability computations. If the full Grundy quotient is finite, its kernel is necessarily nontrivial. The full quotient and the uniform realization of the wild $|P| = 2$ family remain open.

1 Introduction

An abstract finite impartial game can be assembled to realize a prescribed transition table. A finite octal code cannot be assembled heap by heap: its k th digit imposes the same translation-invariant move on every sufficiently large heap. The natural realization problem is to identify the extra algebraic condition imposed by that uniform tail.

The proof architecture has four layers. Reduced transition tables are deterministic in their option-value sets and forbid self-options. The octal rule turns those facts into an exact recurrence and obstruction calculus for the single-heap quotient word. A finite-exception normal form separates abstract heap realization from strict octal homogeneity. Finally, the trace recurrence can be solved either uniformly, as for the tame family, or prefix by prefix with all multiplicities retained, as for Grundy’s game. The main results are the following.

Theorem A. 1. *A reduced valid transition table assigns at most one value to each option-value set. A finite octal code realizes a finite reduced bipartite monoid exactly when its induced heap trace is total, lies in such a table, and generates the monoid.*

2. No table transition has its value among its option values. Thus every whole-, one-, and two-remainder octal digit imposes an explicit inequality on the heap trace.
3. If the code has no split bits and its quotient is finite, the heap trace is ultimately periodic. Exact realization by a fixed no-split code is therefore decidable. If a split trace is already ultimately periodic, its infinite tail is likewise certified by finitely many records.
4. For an arbitrary finite partial decoder, fixed-code split totality is a unary Boolean-language-equation problem and is Π_1^0 -complete already for 0.7. The hardness construction is not a valid misère transition table; validity is the exact remaining separator.
5. Every nontrivial finite quotient has an exact finite-exception unary-heap normal form: finitely many source-local prefix rules followed by the uniform tail $H_n \rightarrow H_{n-1}$. For quotients of order greater than two, its inert pad rules out a verbatim finite-octal encoding of the chosen prefix records.
6. For every $n \geq 2$, the code $0. \underbrace{33 \cdots 3}_{2^{n-1} \text{ digits}}$ has exact quotient $\mathcal{T}_n$. Thus every tame quotient in the finite $|P| = 2$ classification has a finite-code realization.
7. In misère Grundy's game the exact quotients generated through heaps 12, 13, and 18 have orders 6, 12, and 24, respectively, with the presentations and reduced translation tables given below. Any finite full Grundy quotient has a nontrivial kernel.

The first four items give exact algebraic, finite-certificate, and computability boundaries; the fifth is a realization theorem in a comparison class, not for finite octal codes; the sixth is a uniform octal realization theorem; and the seventh combines exact prefix theorems with a conditional structure theorem. None asserts that valid split traces are automatically periodic or that the full Grundy quotient is finite. Sections 2 and 3 build the trace calculus, Section 4 gives the comparison normal form, Section 5 realizes the tame family, and Section 6 gives the Grundy extensions. Section 7 states the formal and open boundaries.

2 Transition tables and deterministic values

Let A be a set of finite impartial games closed under options and disjunctive sum. Under misère play the terminal game is an N -position. For $G, H \in A$, put

$$G \sim_A H \iff o^-(G + X) = o^-(H + X) \text{ for every } X \in A.$$

The quotient $Q(A) = A/\sim_A$ is a commutative monoid. It comes with the distinguished subset P of previous-player wins. The pair (Q, P) is *reduced*: if $x \neq y$, then some z has exactly one of xz, yz in P . Also $1 \notin P$, because the terminal position is N .

Plambeck and Siegel developed this quotient method and the generalized mex rule in [5]; Siegel subsequently proved the abstract classification by valid transition tables in [7]. A transition pair is $(x, E) \in Q \times \text{Pow}(Q)$, with product

$$(x, E)(y, F) = (xy, yE \cup xF). \tag{1}$$

A transition table T on (Q, P) is valid when it is complete, closed under the displayed product, parity-correct,

$$(x, E) \in T \implies x \in P \iff E \neq \emptyset \text{ and } E \cap P = \emptyset, \tag{2}$$

and well-founded: there is $R : Q \rightarrow \mathbb{N}$, with $R(1) = 0$, such that every x has some $(x, E) \in T$ satisfying $R(e) < R(x)$ for all $e \in E$. Siegel's theorem says that a reduced (Q, P) with $1 \notin P$ is an abstract finite misère quotient if and only if it admits such a table.

The natural problem begins after that theorem: a code must generate its table through one translation-invariant heap rule.

2.1 Option-set determinism

The first observation converts the natural problem from a nondeterministic labeling problem into a trace recurrence.

Theorem 2.1 (Option-set determinism). *Let T be a valid transition table on a reduced bipartite monoid (Q, P) . If*

$$(x, E) \in T \quad \text{and} \quad (y, E) \in T,$$

then $x = y$.

Proof. Assume $x \neq y$. Among all contexts separating x and y , choose z with $R(z)$ least. Take a descending pair $(z, F) \in T$. Suppose, after interchanging x, y if necessary, that $xz \in P$ and $yz \notin P$. Closure gives

$$(xz, zE \cup xF), \quad (yz, zE \cup yF) \in T.$$

Parity for xz says that $zE \cup xF$ is nonempty and contains no element of P . The second option set is nonempty as well: a witness coming from zE is unchanged, while a witness xf is replaced by yf . Since $yz \notin P$, parity therefore gives some P -element in $zE \cup yF$. It cannot lie in zE , which is already contained in the P -free first option set. Hence $yf \in P$ for some $f \in F$, while $xf \notin P$. Thus f also separates x, y , but $R(f) < R(z)$, a contradiction. $\square$

Equivalently, every valid table defines a partial function

$$\mu_T(E) = x \iff (x, E) \in T. \tag{3}$$

This strengthening of the option-image principle is purely algebraic: it does not assume that T was first obtained from games. The theorem and its least-rank proof are kernel-checked in the Lean module `Ogdoad.MisereTransition`.

We also record the standard local obstruction in a form suited to heap traces. Define the maximal set

$$M_x = \{y \in Q : \text{there is no } z \text{ with } xz, yz \in P\}.$$

Lemma 2.2 (Maximal obstruction). *Every $(x, E) \in T$ satisfies $E \subseteq M_x$.*

Proof. If $e \in E$ and $xz, ez \in P$, choose any table pair (z, F) . Closure puts $(xz, zE \cup xF)$ in T , and $ez \in zE$ is a P -valued option of the P -value xz , contradicting parity. $\square$

The fact needed to exclude self-options is stronger than reducedness alone; validity supplies it.

Proposition 2.3 (P -reaching contexts and loop-free records). *Let T be a valid transition table on a nontrivial reduced bipartite monoid (Q, P) . For every $x \in Q$ there is a $z \in Q$ with $xz \in P$. Consequently every $(x, E) \in T$ satisfies $x \notin E$.*

Proof. Choose a descending record (x, E) . If $E = \emptyset$, determinism against the terminal record $(1, \emptyset)$ gives $x = 1$. Nontrivial reducedness implies $P \neq \emptyset$, so any $p \in P$ is a P -reaching context. Suppose $E \neq \emptyset$. If $x \in P$, take $z = 1$; if $x^2 \in P$, take $z = x$. Otherwise closure puts (x^2, xE) in T . Its option set is nonempty, so parity and $x^2 \notin P$ force $xe \in P$ for some $e \in E$. Thus every value has a P -reaching context.

The meximal obstruction gives $E \subseteq M_x$. If $x \in E$, then $x \in M_x$, contradicting a context z for which both copies of xz lie in P . $\square$

The one-element quotient has only the terminal record and satisfies the same no-self conclusion trivially. Reducedness without a valid table would not be enough for the P -reaching statement.

3 Exact octal traces

3.1 The trace criterion

Write a finite octal code as three masks. Removing k counters may take a whole heap when C_k holds, may leave one nonempty heap when A_k holds, and may leave two nonempty heaps when B_k holds. These are respectively the 1, 2, 4 bits of the k th octal digit. Let d be the last nonzero digit, put $x_0 = 1$, and seek $x_n = \Phi(H_n)$ for $n \geq 1$. The quotient image of the option set of a heap is exactly

$$E_n = (\{1\} \text{ if } C_n) \cup \{x_{n-k} : A_k, k < n\} \cup \{x_i x_{n-k-i} : B_k, k \leq n-2, 1 \leq i \leq (n-k)/2\}. \quad (4)$$

Equal parts are permitted in an octal split.

Proposition 2.3 immediately gives three absolute exclusions for every exact trace:

$$C_n \implies x_n \neq 1, \quad (5)$$

$$A_k, k < n \implies x_n \neq x_{n-k}, \quad (6)$$

$$B_k, i, j \geq 1, i + j = n - k \implies x_n \neq x_i x_j. \quad (7)$$

These are not bounded-search heuristics: the indicated value occurs in E_n , so equality would make the heap record a forbidden self-option. For example, if the tail has period p , no active one-remainder offset can be a multiple of p ; applying the second exclusion sufficiently far into the tail gives an immediate contradiction.

Theorem 3.1 (Exact trace criterion). *Fix a finite octal code Γ and a finite reduced bipartite monoid (Q, P) . Then $Q(\Gamma) \cong (Q, P)$ if and only if there are a valid table T on (Q, P) and a sequence $(x_n)_{n \geq 0}$ such that*

1. $x_0 = 1$;
2. $(x_n, E_n) \in T$ for every $n \geq 1$, with E_n given by the displayed octal-trace formula;
3. the elements x_n generate Q .

For fixed T and Γ , the sequence is unique whenever it exists.

Proof. Necessity follows by taking the actual transition algebra and the quotient images of the heaps. Conversely define Φ multiplicatively on heap multisets. The transition pair of any position

is the product of its heap pairs. Closure of T therefore puts $(\Phi(G), \Phi''G)$ in T for every position G . Induction on total counter count, using parity, gives

$$o^-(G) = P \iff \Phi(G) \in P.$$

Generation makes Φ onto. Equal images are indistinguishable by the displayed equivalence; distinct images are distinguished by a context in Q , and surjectivity realizes that context by a position. Thus the fibers are exactly the indistinguishability classes. Uniqueness is Theorem 2.1 applied recursively. $\square$

This theorem identifies the general unresolved core precisely: a finite alphabet quadratic-convolution recurrence must remain defined forever. It also separates exact certificates from bounded signatures.

3.2 Finite-state and periodic traces

Suppose every digit lies in $\{0, 1, 2, 3\}$, so B_k is always false. For $n > d$, whole-heap removal is also inactive, and

$$E_n = \{x_{n-k} : A_k, 1 \leq k \leq d\}. \quad (8)$$

Theorem 3.2 (Automatic periodicity). *Let Γ be a finite no-split octal code with last nonzero digit d . If $Q(\Gamma)$ is finite, then its single-heap quotient word $x_n = \Phi(H_n)$ is ultimately periodic. A repeated state occurs among at most $|Q|^d + 1$ consecutive d -windows after the exceptional prefix.*

Proof. The ordered window

$$W_n = (x_{n-d}, \dots, x_{n-1}) \in Q^d$$

determines E_n by the preceding formula. By Theorem 2.1, E_n determines x_n . Thus W_n determines W_{n+1} . The deterministic map acts on at most $|Q|^d$ states, so one state repeats and the entire subsequent orbit is periodic. $\square$

Corollary 3.3. *For a fixed finite no-split code and a fixed finite reduced (Q, P) , exact realization is decidable. The same is true when the code length is bounded in advance.*

Indeed there are finitely many transition tables on Q ; validity is a finite condition, and each induced trace either fails or enters a cycle. Generation and parity can then be checked on finite data. The existential problem over codes of unbounded length is not settled by this argument.

Splitting is the exact place where finite memory fails. The new term

$$S_m = \{x_i x_j : i, j \geq 1, i + j = m\}$$

depends on an unbounded convolution of the prefix. Nevertheless an already periodic trace remains finitely certifiable.

Proposition 3.4 (Periodic convolution). *Assume $x_{n+p} = x_n$ for all $n \geq N$, where $N, p \geq 1$. Then*

$$S_{m+p} = S_m \quad (m \geq 2N + p).$$

Consequently the records (x_n, E_n) of a length- d octal code are periodic for all $n \geq 2N + p + d$.

Proof. For a pair $i + j = m + p$, one of i, j is at least $N + p$; subtract p from that component and use periodicity. Conversely, for $i + j = m$, one component is at least N ; add p to it. Commutativity permits reordering. The one-remainder terms are easier, and whole-heap bits are inactive past d . $\square$

More explicitly, if the code is supported through digit d , then checking one full period of records after any bound $B \geq 2N + p + d$ certifies every later heap record. Thus a proposed preperiod, period, valid table, finite trace through that first tail period, and generation check form an exact infinite-rule certificate. This is a transition-table form of the periodicity argument in [6, 5].

3.3 The Boolean-language boundary

The convolution obstruction can be made exact. For languages $L, M \subseteq \mathbb{N}$, write

$$L + M = \{i + j : i \in L, j \in M\}.$$

Fix a candidate word $x = (x_n)$ and put $X_q = \{n : x_n = q\}$. For each $q \in Q$, let A_q be the language of heap sizes at which q occurs as an option value. In terms of the three masks it is

$$A_q = C_q \cup \bigcup_{k \in A} (k + (X_q \cap \mathbb{N}_{>0})) \cup \bigcup_{k \in B} \bigcup_{uv=q} (k + (X_u \cap \mathbb{N}_{>0}) + (X_v \cap \mathbb{N}_{>0})), \quad (9)$$

where $C_1 = C$ is the finite whole-removal language and $C_q = \emptyset$ for $q \neq 1$. Bounds excluding empty remainders are understood in the shifted terms. For $S \subseteq Q$, define the exact Boolean cell

$$D_S = \bigcap_{q \in S} A_q \cap \bigcap_{q \notin S} (\mathbb{N} \setminus A_q). \quad (10)$$

Proposition 3.5 (Exact language equations). *Let $\mu : \text{Pow}(Q) \rightarrow Q$ be any finite partial decoder. A word x is its total octal trace exactly when*

$$X_q = \bigcup_{\mu(S)=q} D_S \quad (q \in Q) \quad (11)$$

and the failure language

$$F_\mu = \bigcup_{\mu(S) \text{ undefined}} D_S \quad (12)$$

is empty. Coefficient n belongs to F_μ exactly when the decoder has no value on the complete option set at n .

Proof. The first displayed equation is simply the whole-, one-, and two-remainder definition sorted by quotient value. Hence $n \in D_S$ if and only if the complete option-value set at H_n is exactly S . Applying μ coefficientwise gives both displayed conclusions. $\square$

These equations are unary, but they are not finite-state equations. Unary conjunctive grammars already generate nonregular languages such as $\{a^{4^r} : r \geq 0\}$ [3]. In fact the unconstrained trace problem reaches the full effective boundary.

Theorem 3.6 (Arbitrary-decoder hardness). *For the single fixed octal code 0.7, the following problem is Π_1^0 -complete: given a finite commutative monoid M and an arbitrary finite partial decoder $\mu : \text{Pow}(M) \rightarrow M$, decide whether the trace starting from $x_0 = 1$ is total.*

Proof. Non-totality has a least failing coefficient, so it is recursively enumerable and totality is in Π_1^0 .

For hardness, start with a unary unambiguous conjunctive grammar. Emptiness for these grammars is undecidable by an effective reduction from Turing machine emptiness [4, Lemma 12].

Effectively put the grammar in epsilon-free binary conjunctive normal form and prefix a fixed two-letter word, so that lengths 0 and 1 are irrelevant while emptiness is preserved. Unambiguity is needed only for the cited source theorem; the normalization need not preserve it. Let $\mathcal{N}$ be the resulting finite nonterminal set and let $V_n \subseteq \mathcal{N}$ be its truth vector on a^n . Take the full, effectively known Boolean cube $C = \text{Pow}(\mathcal{N})$ as the color set and form the commutative monoid

$$M_C = \{\text{multisets on } C \text{ of degree at most } 2\} \sqcup \{\perp\}.$$

Multiplication is multiset union until degree 2 and is $\perp$ after overflow, with $\perp$ absorbing; the empty multiset is the identity. This is associative: for three ordinary factors both parenthesizations give their union when its total degree is at most two and $\perp$ otherwise, while an existing $\perp$ remains absorbing.

Assign H_h the singleton marker $[V_{h+1}]$. Under 0.7, the option set of H_h for $h \geq 3$ is exactly

$$\{[V_h]\} \cup \{[V_{i+1}, V_{j+1}] : i, j \geq 1, i + j = h - 1\}. \quad (13)$$

The degree-two markers encode every binary cut of the word of length $h + 1$ whose two parts have length at least two. The degree-one predecessor marker, together with the fixed vector V_1 , encodes the two cuts having a part of length one. Because the alphabet is unary, unordered pairs lose no cut information. Thus the exact option set determines V_{h+1} .

More explicitly, the normalized grammar gives an effective finite update map

$$H : C \times \text{Pow}(\{[u, v] : u, v \in C\}) \longrightarrow C.$$

It computes the next truth vector from the predecessor vector and the set of unordered binary-cut vectors; the two cuts involving V_1 are folded into the update. For every decodable signature

$$S = \{[c]\} \cup R, \quad R \subseteq \{[u, v] : u, v \in C\},$$

define $\mu(S) = [H(c, R)]$ when the start nonterminal is absent from $H(c, R)$, and leave $\mu(S)$ undefined when it is present. This is a uniform finite definition, independent of which signatures the eventual trace reaches. Malformed signatures may be left undefined. The genuinely exceptional H_1 cell computes V_2 directly. The H_2 cell is the decodable signature $(c, R) = (V_2, \emptyset)$, for which $H(V_2, \emptyset) = V_3$ because only the two length-one cuts occur. Both cells are left undefined exactly when their output start bit is true.

Before the first accepted length, induction now gives precisely the asserted markers. At the first accepted length the decoder fails. Hence the trace is total if and only if the grammar language is empty. Turing-machine emptiness is Π_1^0 -complete, completing the reduction. $\square$

The decoder in Theorem 3.6 is deliberately arbitrary. It need not extend to a reduced, closed, parity-correct, ranked misère transition table. Therefore the theorem neither proves undecidability of natural realization nor decides a fixed valid-table trace. It proves the more surgical statement needed here: any decision argument must use the valid-table axioms in a load-bearing way. Finiteness of Q , commutativity of its multiplication, and unary input alone are insufficient.

4 A finite-exception heap normal form

The abstract table theorem does admit a uniform one-species heap realization once the finite prefix is allowed to be source-local. This comparison result pinpoints why the direct construction does not immediately become octal.

Theorem 4.1 (Finite-exception unary-heap normal form). *Let (Q, P) be a nontrivial finite reduced bipartite commutative monoid. Suppose T is a closed parity-correct transition table and $R : Q \rightarrow \mathbb{N}$, with $R(1) = 0$, such that every $q \in Q$ has a record $(q, E_q) \in T$ satisfying $R(e) < R(q)$ for all $e \in E_q$. Then there are N and a finite-description impartial ruleset on numerical heaps $H_1, H_2, \dots$ such that*

1. *for $n \leq N$, every option of H_n is a single heap H_j with $j < n$;*
2. *H_N is terminal;*
3. *for every $n > N$, $\text{opts}(H_n) = \{H_{n-1}\}$; and*
4. *the exact misère quotient of all finite sums of these heaps is isomorphic to (Q, P) .*

Proof. First find a star record. Choose a nonidentity value a of least rank and a descending record (a, E) . Every member of E has smaller rank, hence is 1. The set E cannot be empty, since determinism against $(1, \emptyset)$ would give $a = 1$. Therefore $(a, \{1\}) \in T$.

For each $q \in Q$, choose a descending record (q, E_q) . Enumerate the elements of Q in nondecreasing rank and make one prefix heap B_q with options B_e for $e \in E_q$. Every target is an earlier heap. Append one inert padding heap I , followed by heaps $U_0, U_1, \dots$ with

$$\text{opts}(U_0) = \{I\}, \quad \text{opts}(U_{k+1}) = \{U_k\}.$$

Assign the values

$$\Phi(B_q) = q, \quad \Phi(I) = 1, \quad \Phi(U_k) = a^{k+1}.$$

Closure of T and $(a, \{1\}) \in T$ give by induction

$$(a^{k+1}, \{a^k\}) \in T,$$

so every single-heap record lies in T . The transition record of a sum is the product of its component records and therefore also lies in T . All moves strictly lower the sum of the displayed heap ranks. Induction on that rank and parity show

$$o^-(G) = P \iff \Phi(G) \in P$$

for every position G . Each q is represented by B_q , so Φ is surjective. Equal values are indistinguishable, while reducedness and surjectivity realize a separating context for unequal values. Hence the fibers of Φ are exactly the indistinguishability classes. $\square$

This is an eventually translation-invariant finite-exception heap normal form, not a finite-octal realization. Turning a prefix edge $H_i \rightarrow H_{i-d}$ into an octal digit at offset d also creates translated moves from every larger heap. The construction supplies no values for those additional targets. Controlling that cross-talk—possibly by a different, generator-level encoding—remains an unresolved bridge. The comparison class and finite octal games are therefore not ordered by inclusion: the former permits arbitrary source-local prefix moves but only a unary tail, whereas the latter permits persistent multi-removal and split moves.

The inert pad is not merely an expositional nuisance.

Proposition 4.2 (Inert-pad obstruction). *Assume $|Q| > 2$, and choose one strictly descending record (q, E_q) for each $q \in Q$ as in Theorem 4.1. There is no finite octal code with prefix heaps B_q followed by a later terminal heap I such that every B_q has exact option-value set E_q .*

Proof. Determinism identifies every chosen empty record with $(1, \emptyset)$ and identifies all chosen records with option set $\{1\}$ with the unique star value. Since $|Q| > 2$, some chosen E_q therefore contains an $e \neq 1$.

A whole-heap-removal bit contributes only 1, so the option e at B_q must come from a one-remainder or two-remainder bit at some removal size k . Such a bit is persistent. If it leaves one remainder at the earlier heap B_q , then $k < |B_q| < |I|$ and it leaves a remainder at I . If it leaves two remainders at B_q , then $k + 2 \leq |B_q| < |I|$ and it also gives a split option at I . Either conclusion contradicts that I is terminal. $\square$

Thus no placement trick, including spacing the representatives farther apart, turns the proof of Theorem 4.1 verbatim into an octal proof. This does not rule out octal universality: a successful compiler may replace the inert pad by an active bridge or may encode quotient generators contextually rather than representing every chosen descending record by one heap.

5 Uniform realization of the tame family

For $n \geq 2$, let

$$m = 2^{n-1}, \quad L = m + 1,$$

and let Γ_n be the finite octal code

$$\Gamma_n = 0.\underbrace{33 \cdots 3}_{m \text{ digits}}.$$

A move subtracts any number in $\{1, \dots, m\}$ from one heap, deleting it if the remainder is zero and otherwise leaving the one remainder heap. There are no splits.

Define

$$Q_n = \{1, a\} \sqcup \{z_v : v \in \mathbb{F}_2^n\}, \quad P_n = \{a, z_0\}.$$

Identify $v \in \mathbb{F}_2^n$ with an integer in $[0, 2^n)$ and write vector addition as XOR. Multiplication is

$$a^2 = 1, \quad z_u z_v = z_{u \text{ xor } v}, \quad a z_v = z_{v \text{ xor } 1}. \quad (14)$$

This is Siegel's tame quotient $\mathcal{T}_n$: its kernel is $\{z_v\} \cong (\mathbb{Z}/2)^n$, and $|Q_n| = 2^n + 2$ [5, 6].

Theorem 5.1 (Tame-family realization). *For every $n \geq 2$, the exact misère quotient of Γ_n is $\mathcal{T}_n$.*

Proof. For a heap H_h , let $r(h)$ be the residue of h modulo L and define

$$\Phi(H_h) = \begin{cases} 1, & r(h) = 0, \\ a, & r(h) = 1, \\ z_{r(h)}, & 2 \leq r(h) \leq m. \end{cases} \quad (15)$$

Extend Φ multiplicatively.

We first prove the outcome criterion directly. Discard residue-zero heaps. Let $t \in \mathbb{F}_2$ be the parity of the number of residue-one heaps, and let R be the multiset of residues in $\{2, \dots, m\}$. From the displayed multiplication laws,

$$\Phi(X) \in P_n \iff \begin{cases} t = 1, & R = \emptyset, \\ t \text{ xor } \bigoplus_{r \in R} r = 0, & R \neq \emptyset. \end{cases} \quad (16)$$

Induct on total counter count. A legal move changes one residue r to a different residue s . If $R = \emptyset$ and $t = 1$, changing a residue 0 or 1 either flips the parity or creates a nonzero XOR, so every option fails this criterion. If $R \neq \emptyset$ and the displayed XOR is zero, there must be at least two residues at least 2: one such residue cannot cancel either 0 or 1. A move changes the XOR by $r \text{ xor } s \neq 0$, and at least one large residue remains. Hence every option of a predicted P -position is predicted N .

Conversely suppose the position is predicted N . If $R = \emptyset$ and $t = 0$, move a residue-one heap to residue zero when one exists. Otherwise the nonempty position has a positive residue-zero heap; subtracting m moves it to residue one. Either move makes $t = 1$.

Now suppose at least two residues at least 2 occur, and write

$$w = t \text{ xor } \bigoplus_{r \in R} r \neq 0.$$

Choose a component residue r whose binary expansion contains the highest set bit of w and put $s = r \text{ xor } w$. Such a component exists among the residues contributing to the XOR: it may be a residue-one component when the highest bit is the low bit, and otherwise it lies in R . Then $s < r$, so subtracting $r - s \leq m$ is legal, the new XOR is zero, and at least one large residue remains. Finally, if exactly one large residue r occurs, move it to residue $1 - t < r$. No large residue remains and the new residue-one parity is odd. This proves the displayed criterion is the actual misère outcome for every position.

The map is onto. The heap residues provide a and $z_2, z_4, \dots, z_m$; also $z_2^2 = z_0$ and $az_0 = z_1$. These are a binary basis for every z_v . Finally (Q_n, P_n) is reduced. Distinct z_u, z_v are separated by z_u . The elements $1, a$ are separated by 1; the context a separates 1 from z_v unless $v = 1$, when z_1 does; and the context 1 separates a from z_v unless $v = 0$, when z_0 does. Outcome correctness, surjectivity, and reduction identify the fibers of Φ with exact indistinguishability classes. $\square$

For $n = 2$, the construction is 0.33 and gives the six-element quotient $\mathcal{T}_2$. The theorem realizes the entire tame arm in Siegel's classification of finite quotients with $|P| = 2$; the other arm consists of the tame extensions of the eight-element wild quotient $\mathcal{R}_8$ [7]. The base $\mathcal{R}_8$ itself is realized by 0.75 [5], but a uniform finite-code realization of all its tame extensions is not supplied here.

6 Exact initial quotients of misère Grundy's game

In Grundy's game a heap may be split into two unequal positive heaps:

$$\text{opts}(H_n) = \{H_i + H_{n-i} : 1 \leq i < n - i\}.$$

The rule preserves total counters but decreases the well-founded quantity $\sum_h \max(h - 2, 0)$. Its misère form is a longstanding difficult example; the early analysis goes back to Grundy and Smith [2] and is discussed in [1].

6.1 The tame prefix through heap 12

We first give an exact transition certificate through heap 12. Write

$$\mathcal{T}_2 = \langle a, b \mid a^2 = 1, b^3 = b \rangle, \quad P = \{a, b^2\}.$$

The heap values are

$$x_1 = x_2 = 1, \quad x_n = \begin{cases} a, & n \equiv 0 \pmod{3}, \\ 1, & n \equiv 1 \pmod{3}, \\ b, & n \equiv 2 \pmod{3} \end{cases} \quad (3 \leq n \leq 12).$$

Their distinct transition records are

$$(1, \emptyset), (a, \{1\}), (1, \{a\}), (b, \{1, a\}), \\ (a, \{1, b\}), (1, \{a, b\}), (b, \{1, a, ab\}).$$

Closing these records under the transition product gives the following 24 pairs; each row lists the option sets paired with its left-hand value.

1	$\emptyset; \{a\}; \{a, b\}; \{a, ab\}; \{a, b, ab\}$
b	$\{1, a\}; \{b^2, ab^2\}; \{1, a, ab\}; \{b^2, ab, ab^2\}; \{1, b^2, a, ab\}; \{1, a, ab, ab^2\}; \{1, b^2, a, ab, ab^2\}$
b ²	$\{b, ab\}; \{b, ab, ab^2\}$
a	$\{1\}; \{1, b\}; \{1, ab\}; \{1, b, ab\}$
ab	$\{1, b, a\}; \{b, b^2, ab^2\}; \{1, b, b^2, a\}; \{1, b, a, ab^2\}; \{1, b, b^2, a, ab^2\}$
ab ²	$\{b, b^2, ab\}$

The table is closed and every pair obeys parity; a, b occur as heap values and $\mathcal{T}_2$ is reduced. The trace criterion therefore proves:

Proposition 6.1. *The exact misère quotient generated by $H_1, \dots, H_{12}$ is $\mathcal{T}_2$.*

6.2 The first extension at heap 13

Before extending this quotient, we isolate a substitution point that is easy to miss. Write $G \equiv_- H$ when G and H have the same misère outcome after addition of every finite impartial game, not merely every game in the present heap universe.

Lemma 6.2 (Global substitution). *Let U be the additive closure of games G_i , and suppose that each G_i is globally equivalent to a representative R_i lying in an additively closed subuniverse $V \subseteq U$. Then inclusion induces an isomorphism $Q(V) \cong Q(U)$.*

Proof. Global equivalence is an additive congruence, so every $X \in U$ is globally equivalent to the sum $r(X) \in V$ obtained by replacing its components. If $A, B \in V$ agree against all V -contexts, then for every $X \in U$,

$$o^-(A + X) = o^-(A + r(X)) = o^-(B + r(X)) = o^-(B + X).$$

The converse follows because $V \subseteq U$, and every U -class has a representative in V . $\square$

The Grundy–Smith canonical reduction rules are global reductions. Applied to the explicit option lists on heaps through 12, they give

$$\begin{array}{l|l} 0 & H_1, H_2, H_4, H_7, H_{10} \\ a = *1 & H_3, H_6, H_9, H_{12} \\ b = *2 & H_5, H_8, H_{11}. \end{array}$$

These are the reductions displayed in [1, pp. 419–421]. Consequently adjoining H_{13} cannot secretly refine one of these earlier identifications: they remain valid against the new context H_{13} and all of its sums.

Heap 13 has option-value set $\{a, b, b^2\}$. The alleged continuation $x_{13} = 1$ is impossible: multiplying its transition by two copies of the H_5 transition produces

$$(b^2, \{b, b^2, ab, ab^2\}),$$

which violates parity because both the value and one option lie in P . Concretely $H_4 + 2H_5$ is P , while $H_{13} + 2H_5$ has the P -option $H_8 + 3H_5$ and is N .

The extension can still be solved symbolically. Let c denote the new heap value, with transition $c \rightarrow \{a, b, b^2\}$. For raw monomials $a^e b^j c^k$, direct induction on (k, j, e) gives the following complete outcome law:

$$a^e b^j c^k \in P \iff \begin{cases} k \text{ even, } e \text{ even, } j > 0 \text{ even; or} \\ k \text{ even, } e \text{ odd, } j = 0; \text{ or} \\ k \text{ odd, } e \text{ odd, } j = 0 \text{ or } j \geq 3 \text{ odd.} \end{cases} \quad (17)$$

For completeness, the induction uses only

$$a \rightarrow 1, \quad b \rightarrow \{1, a\}, \quad c \rightarrow \{a, b, b^2\}.$$

Here is the complete case induction. Put $\epsilon = e \bmod 2$. For a listed P -state with k even, either $(\epsilon, j) = (0, \text{positive even})$ or $(1, 0)$; the a, b, c moves land respectively in the complementary parity classes, and the three c replacements land at $(1, j), (0, j+1), (0, j+2)$ with k odd. None is listed. For a listed state with k odd, $\epsilon = 1$ and $j = 0$ or odd at least 3; the a move sets $\epsilon = 0$, the b moves make j even, and the c moves land at $(0, j), (1, j+1), (1, j+2)$ with k even. Again none is listed.

Conversely, every unlisted nonterminal state has the following listed option. When k is even:

ϵ	j	move to a P -state
0	0	$c \rightarrow a$ (unless $k = 0$, when the position is terminal)
0	1	$b \rightarrow a$
0	odd ≥ 3	$b \rightarrow 1$
1	positive even	$a \rightarrow 1$
1	1	$b \rightarrow 1$
1	odd ≥ 3	$b \rightarrow a$

When k is odd, the witnesses are

ϵ	j	move	ϵ	j	move
0	0	$c \rightarrow a$	0	positive even	$c \rightarrow b^2$
0	odd	$c \rightarrow b$	1	1	$b \rightarrow 1$
1	2	$c \rightarrow a$	1	even ≥ 4	$b \rightarrow 1$

The omitted odd- k cases are exactly the listed P cases. This proves the misère recursion for all multiplicities; no exponent cutoff is used.

The outcome law also proves the presentation rather than merely suggesting it. It first factors raw monomials through the sixteen forms $a^\epsilon b^j c^\eta$, with $\epsilon, \eta \in \{0, 1\}$ and $j \in \{0, 1, 2, 3\}$, using $a^2 = c^2 = 1$ and $b^4 = b^2$. It is invariant under $b^2 \sim ab^3c$. The congruence generated by this relation pairs the eight forms in the b^2 -ideal into four two-element orbits and leaves the eight forms with $j = 0, 1$ fixed. Thus it has exactly twelve candidate classes. The resulting quotient has presentation

$$Q_{13} = \langle a, b, c \mid a^2 = c^2 = 1, b^4 = b^2, b^2 = ab^3c \rangle, \quad P_{13} = \{a, ac, b^2\}. \quad (18)$$

It has the twelve normal forms

$$1, c, b, bc, b^2, b^2c, b^3, b^3c, a, ac, ab, abc.$$

Reduction is visible from their distinct P -translation rows:

$1 \{b^2, a, ac\}$	$c \{b^2c, a, ac\}$
$b \{b, b^3\}$	$bc \{bc, b^3c\}$
$b^2 \{1, b^2, abc\}$	$b^2c \{c, b^2c, ab\}$
$b^3 \{b, b^3, ac\}$	$b^3c \{bc, b^3c, a\}$
$a \{1, c, b^3c\}$	$ac \{1, c, b^3\}$
$ab \{b^2c, ab\}$	$abc \{b^2, abc\}$

Here the row at x is $\{z : xz \in P_{13}\}$. All rows differ, so the bipartite monoid is reduced. Thus the all-multiplicity outcome law proves:

Proposition 6.3. *The exact misère quotient generated by $H_1, \dots, H_{13}$ is the twelve-element reduced bipartite monoid in the preceding presentation.*

6.3 The second extension at heap 18

The next heaps have values

$$x_{14} = b, \quad x_{15} = a, \quad x_{16} = c, \quad x_{17} = b.$$

Here these assertions are again global, not guesses made inside Q_{13} . With 0 denoting the terminal game, the split rule gives

$$\begin{aligned} \text{opts}(H_{14}) &= \{c, ab, a, 0\}, & \text{opts}(H_{15}) &= \{c, b, 0\}, \\ \text{opts}(H_{16}) &= \{ac, b^2, b, a\}, & \text{opts}(H_{17}) &= \{c, ab, a, 0\}. \end{aligned}$$

The reversible-option reductions of Grundy and Smith give respectively $H_{14} \equiv_- b$, $H_{15} \equiv_- a$, $H_{16} \equiv_- c$, and $H_{17} \equiv_- b$; these are also the four reductions printed in [1, p. 421]. Heap 18 has exact option set $\{0, b, c, bc\}$ and creates a generator $d = H_{18}$. By Lemma 6.2, every position generated by $H_1, \dots, H_{18}$ is globally equivalent to a count state $a^e b^j c^k d^r$, and every such count state occurs in that universe.

The all-multiplicity recursion extends once more:

$$\begin{aligned} a^e b^j c^k d^r \in P \quad &\Longleftrightarrow \quad (r \text{ even and } a^e b^j c^k \text{ satisfies the} \\ &\text{preceding three-generator law}) \\ &\text{or } (r \text{ odd, } e \text{ even, } j = 0). \end{aligned} \tag{19}$$

To check the recursion, use $d \rightarrow \{1, b, c, bc\}$. If r is even and the three-generator state is P , the preceding proof handles the a, b, c moves, while every d move has odd r and either odd e or positive j . If the three-generator state is N , use its displayed winning move. The only case with no a, b, c component is terminal when $r = 0$; for positive even r , the move $d \rightarrow 1$ reaches the odd- r clause.

If r is odd, a state in the displayed P clause has even e and $j = 0$. An a move makes e odd, a c move makes either e odd or j positive, and a d move lands, at even r , in one of

$$(\epsilon, j, k) = (0, 0, k), (0, 1, k), (0, 0, k+1), (0, 1, k+1),$$

all of which fail the three-generator criterion. Conversely an odd- r N -state has a P move in the following exhaustive cases:

ϵ	j	move	ϵ	j	move
1	0	$a \rightarrow 1$	0	1	$b \rightarrow 1$
0	even ≥ 2	$d \rightarrow 1$ if k even; $d \rightarrow c$ if k odd	0	odd ≥ 3	$d \rightarrow b$ if k even; $d \rightarrow bc$ if k odd
1	1	$b \rightarrow a$	1	even ≥ 2	$d \rightarrow bc$ if k even; $d \rightarrow b$ if k odd
1	odd ≥ 3	$d \rightarrow c$ if k even; $d \rightarrow 1$ if k odd			

Each target is one of the three even- r cases already proved. This establishes the outcome formula for all $(e, j, k, r) \in \mathbb{N}^4$.

As before, the formula itself identifies the finite presentation. Before the last relation there are 32 forms $a^\epsilon b^j c^\eta d^\rho$, with $\epsilon, \eta, \rho \in \{0, 1\}$ and $0 \leq j \leq 3$, subject to $a^2 = c^2 = d^2 = 1$ and $b^4 = b^2$. The relation $b^2 = ab^3c$ acts freely in the sixteen-element b^2 -ideal, producing eight pairs, and leaves the sixteen forms with $j = 0, 1$ fixed. Hence exactly 24 candidate forms remain, and the outcome law descends to them. The quotient is

$$Q_{18} = Q_{13} \times \langle d \mid d^2 = 1 \rangle, \quad P_{18} = \{a, ac, b^2, d, cd\}. \quad (20)$$

Its 24 normal forms are the twelve above and their products by d . For an explicit reduction certificate, let R be the ordered set of twelve base forms, put $P_0 = \{a, ac, b^2\}$ and $P_1 = \{1, c\}$, and define

$$A_q = \{z \in R : qz \in P_0\}, \quad B_q = \{z \in R : qz \in P_1\}.$$

The first coordinate A_q is the Q_{13} row already displayed. The second coordinates are

q	B_q	q	B_q	q	B_q
1	$\{1, c\}$	c	$\{1, c\}$	b	$\emptyset$
bc	$\emptyset$	b^2	$\emptyset$	b^2c	$\emptyset$
b^3	$\emptyset$	b^3c	$\emptyset$	a	$\{a, ac\}$
ac	$\{a, ac\}$	ab	$\emptyset$	abc	$\emptyset$

The translation row of q in Q_{18} is (A_q, B_q) , while that of qd is (B_q, A_q) . The twelve displayed pairs and their twelve swaps are all distinct, so every two normal forms have a separating context. Hence:

Theorem 6.4 (Exact Grundy prefix). *The exact misère quotient generated by $H_1, \dots, H_{18}$ is the 24-element reduced bipartite monoid in the preceding presentation.*

The statements concern entire submonoids generated by the indicated heaps; no multiplicity bound is present. They do not decide whether the full Grundy quotient is finite. They do show that the apparent period-three tame pattern fails for an exact, explicit reason at heap 13, and they give the first two symbolic extension layers for a future infinite-family or stabilization argument.

Only the individual-heap reductions displayed above are global. The relations $c^2 = d^2 = 1$, $b^4 = b^2$, and $b^2 = ab^3c$ are conclusions inside the localized quotient Q_{18} ; they are not asserted as identities in the global semigroup of all misère impartial games. (The relation $a^2 = 1$ is global.)

6.4 A necessary kernel obstruction to finiteness

The split rule and finiteness alone force a nontrivial recurrent algebraic component. This does not decide finiteness, but it excludes stabilization in any quotient whose kernel is trivial.

Theorem 6.5 (Finite Grundy quotients have nontrivial kernel). *Suppose the full misère quotient Q of Grundy's game is finite. Then there is a heap transition (c, E) with $c^2 \in E$. Every two consecutive positive powers of c are distinct, and the powers of c are eventually periodic with a period of length at least two. In particular, the kernel of the finite commutative monoid Q is nontrivial.*

Proof. Color each positive integer n by the heap value $x_n \in Q$. A standard distinct-summand form of Schur's theorem gives $i < j$ for which

$$x_i = x_j = x_{i+j} = c.$$

For completeness, it follows directly from finite Ramsey: color the edge $\{r, s\}$ by $x_{|r-s|}$ and take a monochromatic K_4 with vertices $r_0 < r_1 < r_2 < r_3$. Splitting the outer difference at r_1 and at r_2 gives two monochromatic Schur triples; their two summands cannot both be equal, since that would force $r_1 = r_2$. Choose the unequal one.

Because $i < j$, the position $H_i + H_j$ is a legal option of H_{i+j} . Consequently $c^2 \in E$. Closure of the transition table puts the r -fold transition power

$$(c^r, c^{r-1}E)$$

in the table for every $r \geq 1$. Its option set contains c^{r+1} . Proposition 2.3 therefore gives $c^r \neq c^{r+1}$ for every $r \geq 1$. Finiteness of Q makes the power sequence eventually periodic, and the displayed inequalities exclude period one.

Let K be the kernel, or minimal ideal, of Q . Since Q is a finite commutative monoid, K is a group; write z for its identity. Surjectivity of the quotient map supplies a transition with value z . Multiplying it by (c, E) gives a kernel-valued transition of value $k = zc$ whose option set contains

$$zc^2 = (zc)^2 = k^2.$$

Again no-self gives $k \neq k^2$. A one-element kernel would have $k = z = k^2$, a contradiction. $\square$

The conclusion is structural rather than contradictory: period two already occurs in tame quotients. To prove the full Grundy quotient infinite, one still needs either unbounded kernel rank or infinitely many distinct translation rows, not merely one nonidentity kernel element.

7 Formal verification and open boundary

The results divide the original problem into sharper pieces.

1. Every tame finite quotient $\mathcal{T}_n$ has a uniform finite-code realization. For the $|P| = 2$ classification, the natural realization question remains for the family obtained by tame extension of $\mathcal{R}_8$.
2. Fix a finite (Q, P) , a valid table T , and a finite code Γ . At stage n , the preceding values determine E_n , and determinism supplies at most one possible x_n . Exact realization through T is therefore precisely totality of this partial recurrence plus generation of Q . For no-split codes this is finite-state. With splitting, the first unresolved problem is deciding totality of the unbounded convolution recurrence for a fixed T and Γ . Theorem 3.6 proves Π_1^0 -completeness when the decoder is arbitrary, so a positive algorithm must exploit validity; the theorem does not transfer hardness to valid tables.

3. Even a decision theorem for every fixed split code would leave a second problem: synthesis ranges over codes of unbounded length. A decidable monoid-level characterization needs a computable length bound or pumping normalization, unless natural realization is instead proved undecidable. There are only finitely many tables on fixed Q , so table enumeration introduces no third gap.
4. For Grundy’s game, Q_{13} and Q_{18} are exact. A solution of the full problem must either continue these presentations uniformly and prove stabilization, or produce an infinite family of distinct translation rows with explicit separating contexts. In the finite branch, Theorem 6.5 now additionally requires a nontrivial kernel and nonconstant eventual power cycle.

The import-only entry point `Ogdoad.Papers.MisereNaturalRealization` exposes the Lean development. It checks transition determinism, meximal inclusion, P -reaching contexts, the no-self-option theorem and all three octal digit exclusions. It also checks the generic rank/parity proof that a surjective multiplicative value map has exactly the quotient fibers claimed, the complete-option-record periodicity bound and finite infinite-tail certificate, and the abstract prefix/pad/unary-tail construction underlying Theorem 4.1. It also checks the inert-pad obstruction, the coefficient-wise unary-language dictionary and failure language, the distinct-summand finite-color bridge (via Mathlib’s Hindman theorem), and the algebraic implication from a square option to a nonconstant eventual power period. The grammar-hardness reduction and kernel deduction are paper-level, as are the transport from typed prefix heaps to their numerical rank-order enumeration, and passage from ordered heap lists to multisets. The tame-family strategy and the Grundy prefix presentations likewise remain paper proofs; their finite monoid models and exponent recursions are the next natural formalization targets.

8 Conclusion

Natural realization is not an additional finite-table axiom. It is a totality problem for a quotient-valued heap trace. Transition-table determinism makes that trace canonical: no-split codes become finite-state, while split codes expose exactly the convolution that prevents the same automatic periodicity argument. The unary-language reduction shows that this is not a superficial failure of the proof: without valid-table structure, the fixed-code totality problem is already Π_1^0 -complete.

The finite-exception normal form shows that a uniform infinite tail alone is not the difficulty: every finite quotient admits such a realization. The octal constraint couples each intended finite-prefix edge to infinitely many translated edges, and the no-self inequalities give the first absolute rejection test for that coupling. The inert-pad theorem rules out the most direct compiler, but leaves active bridges open. What remains is to decide totality under the full valid-table axioms and then control unbounded code synthesis.

Within this framework the tame family is complete. One explicit sequence of finite codes realizes every $\mathcal{T}_n$, with an outcome proof valid for all positions. Grundy’s game exhibits the complementary phenomenon: its tame prefix breaks at heap 13 and extends again at heap 18, but each finite stage remains exactly computable as a reduced all-multiplicity quotient. Any finite limit must have a nontrivial kernel, but that condition is compatible with period two. A complete solution must now control the extension process uniformly—either by eventual stabilization or by explicit separating contexts for infinitely many classes.

References

- [1] Elwyn R. Berlekamp, John H. Conway, and Richard K. Guy. *Winning Ways for Your Mathematical Plays*. A K Peters, Natick, MA, second edition, 2001.
- [2] P. M. Grundy and C. A. B. Smith. Disjunctive games with the last player losing. *Proceedings of the Cambridge Philosophical Society*, 52:527–533, 1956.
- [3] Artur Jež. Conjunctive grammars generate non-regular unary languages. *International Journal of Foundations of Computer Science*, 19(3):597–615, 2008.
- [4] Artur Jež and Alexander S. Okhotin. Unambiguous conjunctive grammars over a one-symbol alphabet. *Theoretical Computer Science*, 665:13–39, 2017.
- [5] Thane E. Plambeck and Aaron N. Siegel. Misère quotients for impartial games. *Journal of Combinatorial Theory, Series A*, 115(4):593–622, 2008.
- [6] Aaron N. Siegel. Misère games and misère quotients, 2006.
- [7] Aaron N. Siegel. The structure and classification of misère quotients, 2007.